

C++ Social Media Application

Technical Specification & System Architecture Guide

OOP

Core Paradigm

STL

Vectors & Stacks

19

Console Features

C++11

Standard

System Overview

The C++ Social Media Console Application is an interactive, text-driven system engineered to simulate core social networking behaviors. Built strictly around Object-Oriented Programming principles and leveraging Standard Template Library (STL) components like vectors and stacks, it offers modular user management, graph-based friendship networks, messaging queues, and robust undo capabilities.

User Profile Management

Enables creation of rich user profiles including demographics like city, institution, multi-item interests, and customizable hobbies with automated unique identifiers.

Networking & Suggestion

Facilitates friend connections, real-time mutual friend calculations, and smart multi-criteria recommendations matched across cities, schools, or shared interests.

Core Modules & Menu Reference

The application is controlled via a centralized command-line interface loop with nineteen distinct operational endpoints designed to manage runtime states efficiently.

Option	Module Function	Implementation Description
1–2	Account & Discovery	Creates new account profiles and lists all active users.
3–6	Social Networking	Follows users, displays friends list, mutuals, and suggests matches.
7–8	Messaging & Switch	Handles live chat loops via stack structures and account switching.
9–14	Groups & Posts	Manages group memberships, news feed posts, comments, and profiles.
15–18	Undo Operations	Utilizes stack frames to restore deleted posts and unfriended users.

Compilation Instructions

Compile using any C++11 compliant compiler via command line: `g++ -std=c++11 main.cpp -o social_app` followed by executing `./social_app`. Ensure standard headers like `<vector>`, `<stack>`, and `<queue>` are linked properly.